

A contingency table

A contingency table cross-classifies counts or probabilities for two categorical variables.

Synthetic survey:

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

Which two variables organize this synthetic survey?

A joint count

Synthetic survey of 100 students.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

How many students use both a tablet and a laptop?

A marginal count

Synthetic survey of 100 students.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

How many students use a laptop, regardless of tablet use?

A marginal probability

Select one student uniformly from this synthetic survey.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

What is the probability of tablet use?

A joint probability

Select one student uniformly from this synthetic survey.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

What is the probability of tablet AND laptop use?

A union probability

Select one student uniformly from this synthetic survey.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

What is the probability of tablet OR laptop use?

Neither category

Select one student uniformly from this synthetic survey.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

What is the probability of neither tablet nor laptop use?

Condition on a row

Select one student uniformly from the tablet users in this synthetic survey.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

What is the probability of laptop use?

Condition on a column

Select one student uniformly from the laptop users in this synthetic survey.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

What is the probability of tablet use?

Different denominators

Select one student uniformly from this synthetic survey.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

Why do $P(\text{laptop} \mid \text{tablet})$ and $P(\text{tablet} \mid \text{laptop})$ have different denominators?

An independence check

Select one student uniformly from this synthetic survey.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

Are tablet use and laptop use independent in these data?

Mutual exclusivity check

Synthetic survey of 100 students.

	Laptop	No laptop	Total
Tablet	18	12	30
No tablet	42	28	70
Total	60	40	100

Are tablet use and laptop use mutually exclusive?

Complete the counts

Synthetic survey table. Every student is in exactly one row and one column.

	Online	In person	Total
Morning	12	?	30
Afternoon	8	22	?
Total	?	40	60

What counts fill the missing cells?

A probability contingency table

Synthetic routes: A is chosen with probability 0.4, B with 0.6. $P(\text{delay} \mid A) = 0.1$; $P(\text{delay} \mid B) = 0.2$.

What joint probabilities belong in a route-by-delay table, including its margins?

Conditioning on a combined category

Synthetic counts, with each job in one cell.

Route	Delayed	On time
A	4	36
B	12	28
C	10	10

Given a delayed job, what is the probability it used route A or B?

Percent of which group?

Synthetic survey: 18 of 30 tablet users also use laptops; 18 of 60 laptop users also use tablets.

Which statement corresponds to 60%?

A dependent table

Select one student uniformly from this synthetic survey.

	Laptop	No laptop	Total
Tablet	24	6	30
No tablet	36	34	70
Total	60	40	100

Does tablet use change the probability of laptop use?

Scope of the table

A convenience survey finds laptop use and tablet use independent among respondents.

Does this establish independence for all students at the college?